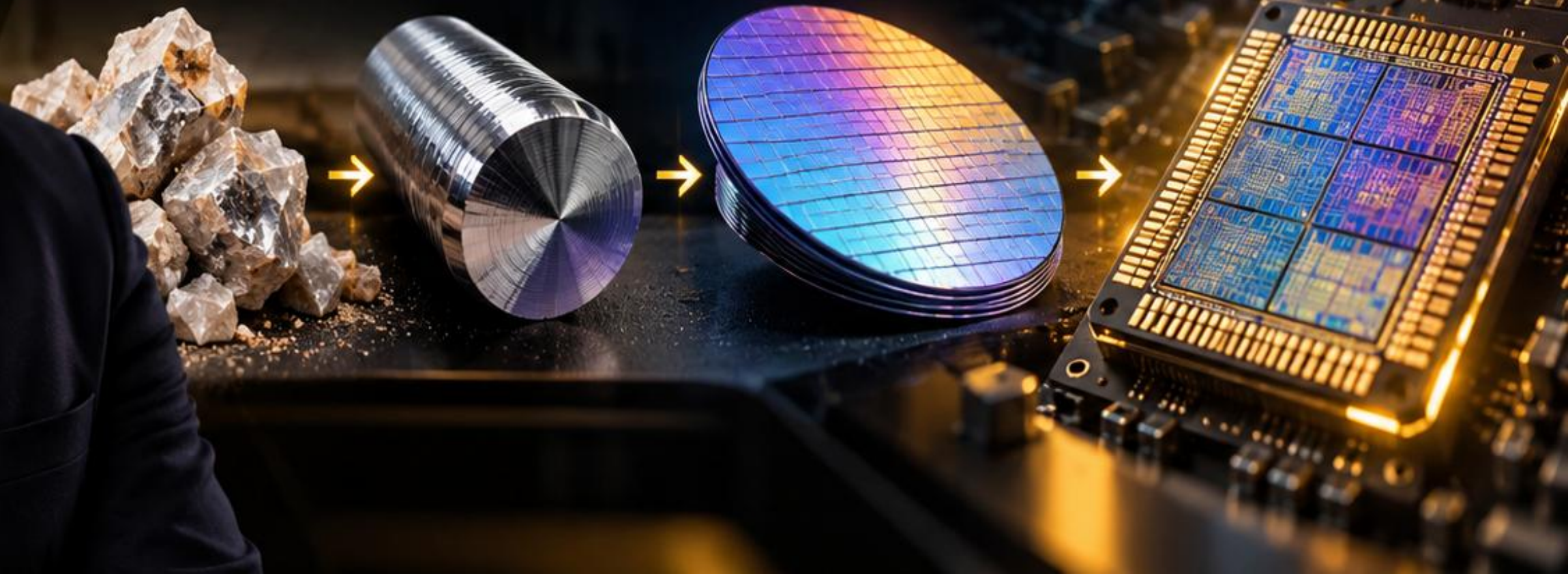




How Are Computer Chips Actually Made? From Sand to a ₹₹₹ Chip

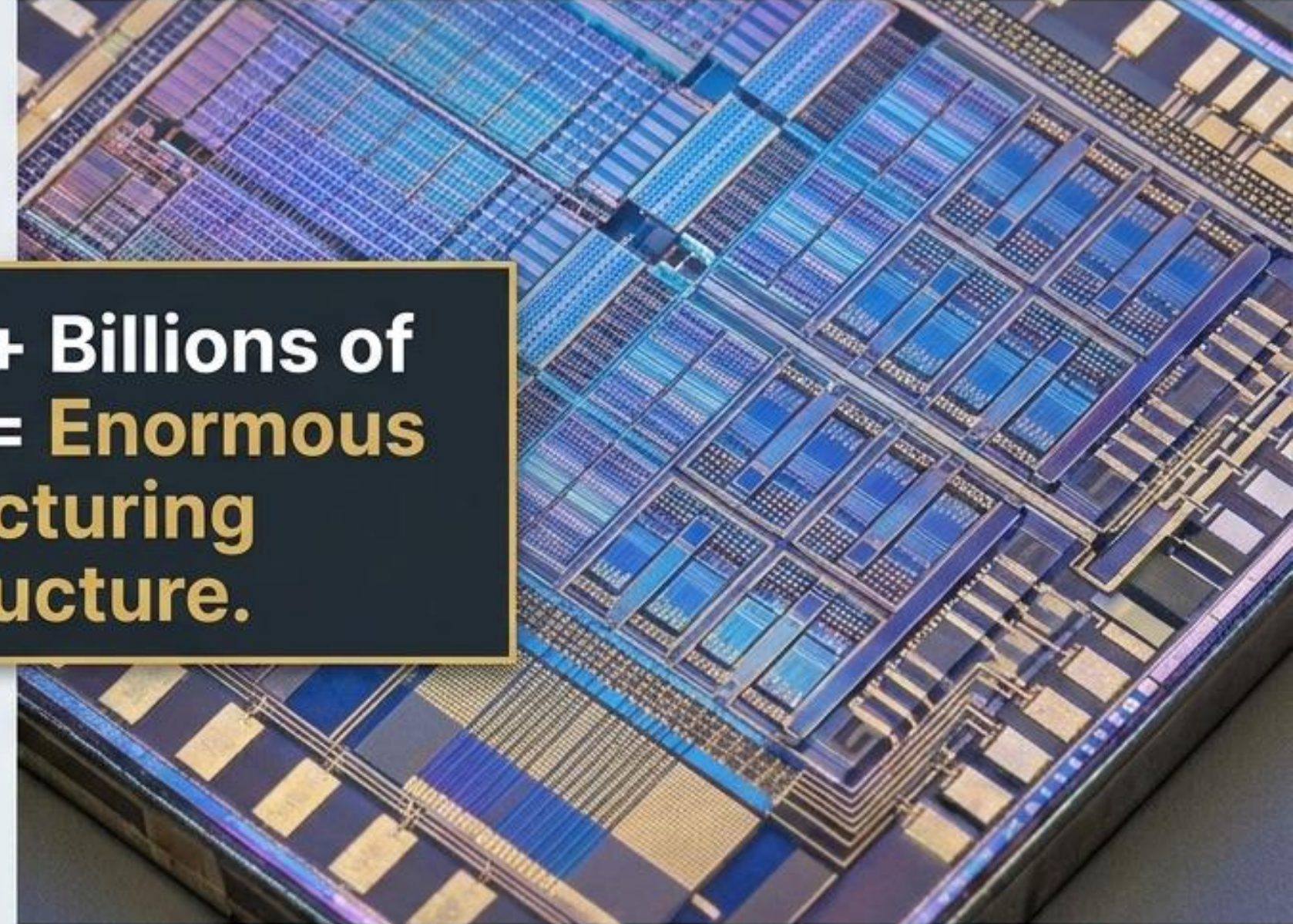
THE EXTRAORDINARY JOURNEY BEHIND
THE TECHNOLOGY THAT POWERS OUR WORLD

Daksh Khandelwal
Founder, ThinkHub



How Are Computer Chips Actually Made?

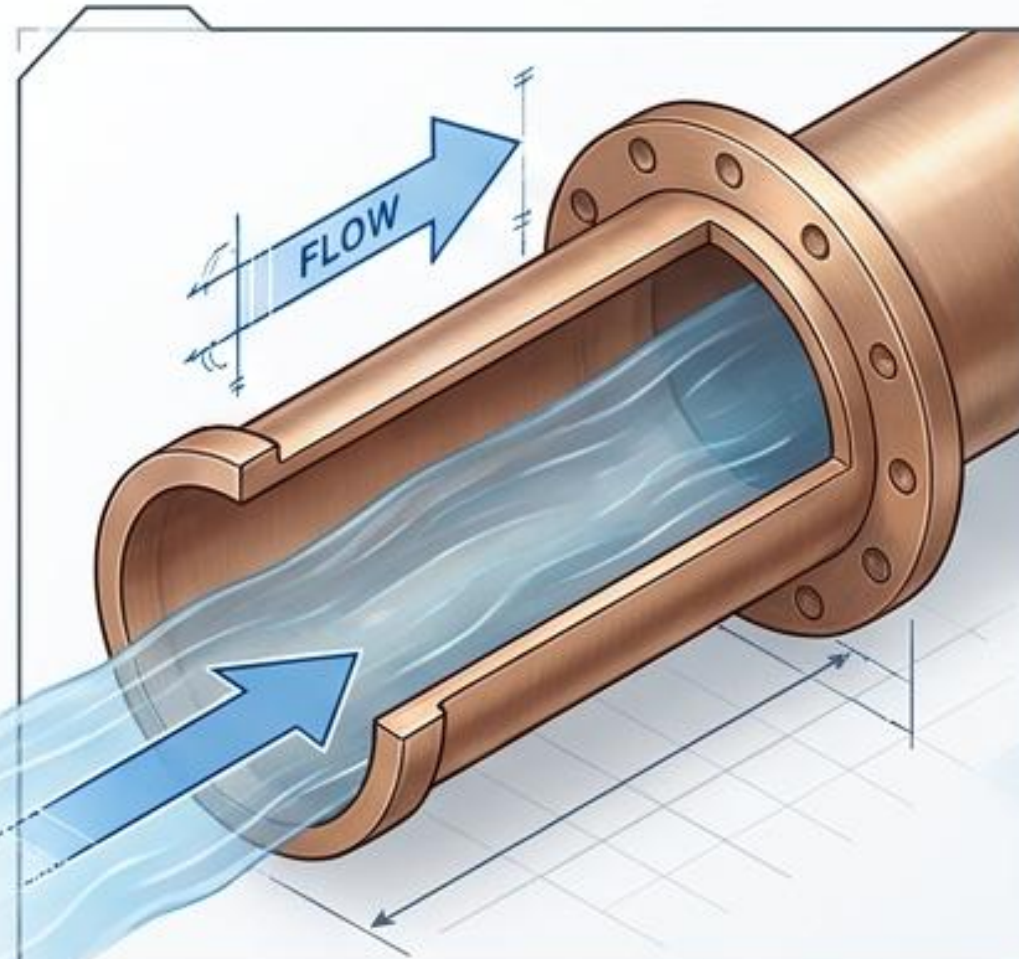
From Sand to a ₹₹₹ Chip



Tiny Space + Billions of Transistors = Enormous Manufacturing Infrastructure.

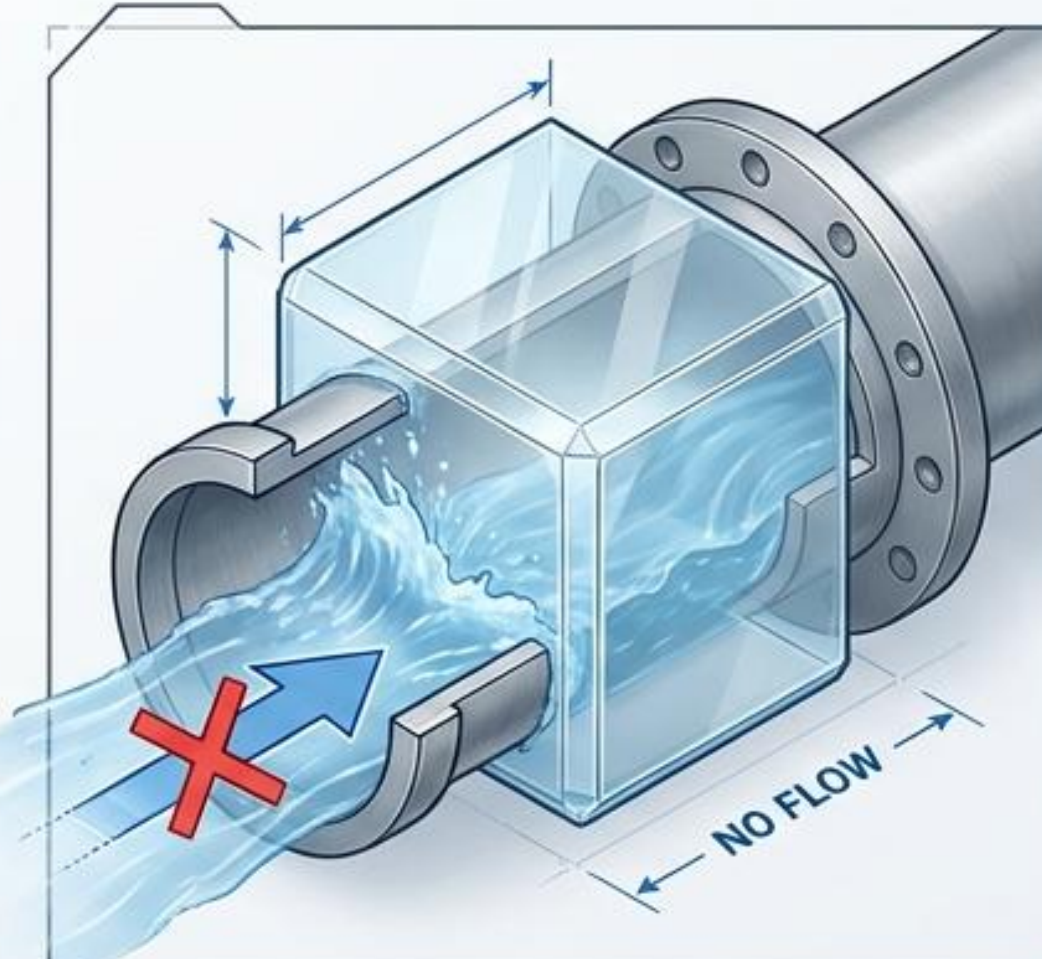
A computer chip is not carved from sand. It is built through an extraordinarily precise sequence of chemical, physical, optical, and electrical processes.

What Exactly Is a Semiconductor?



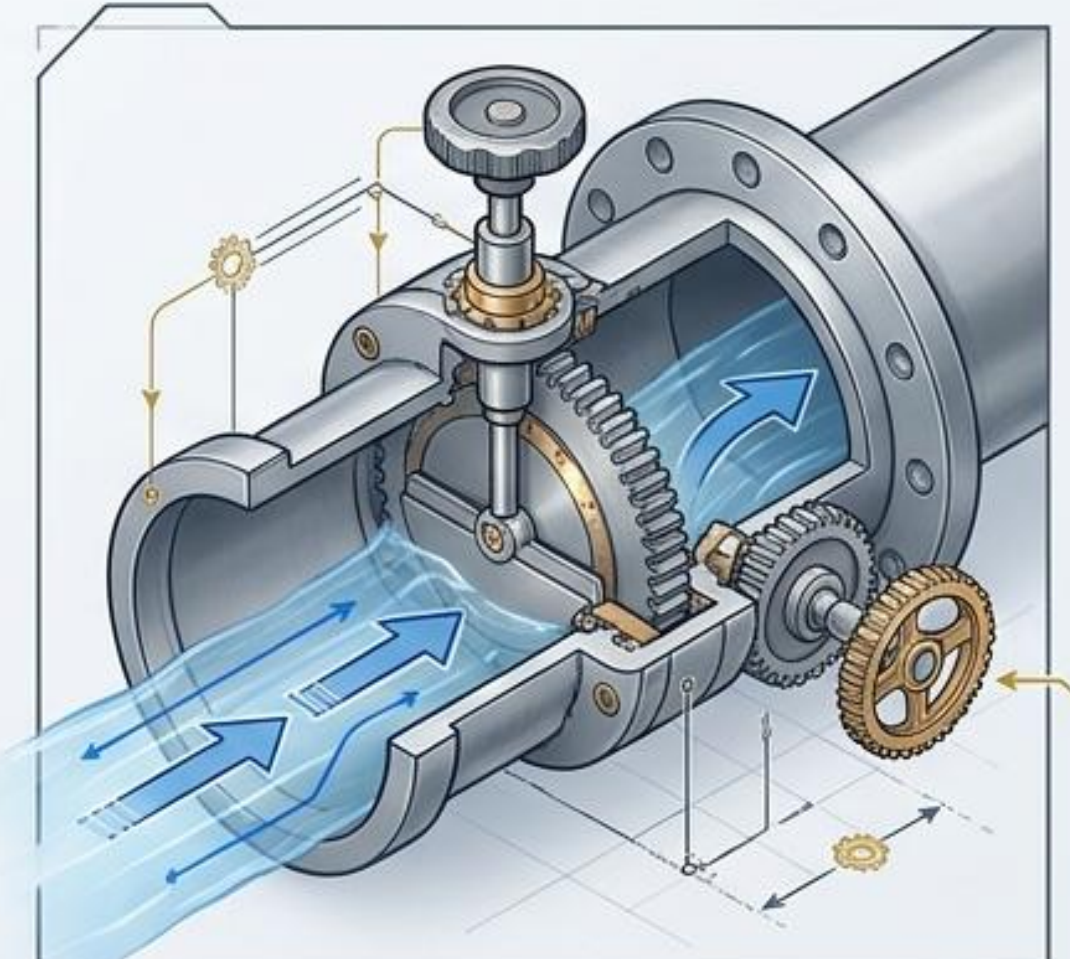
Conductor (Copper)

Allows electric current to flow easily.
Used for wires and interconnects.



Insulator (Glass)

Resists the flow of electric current.
Used to separate structures.

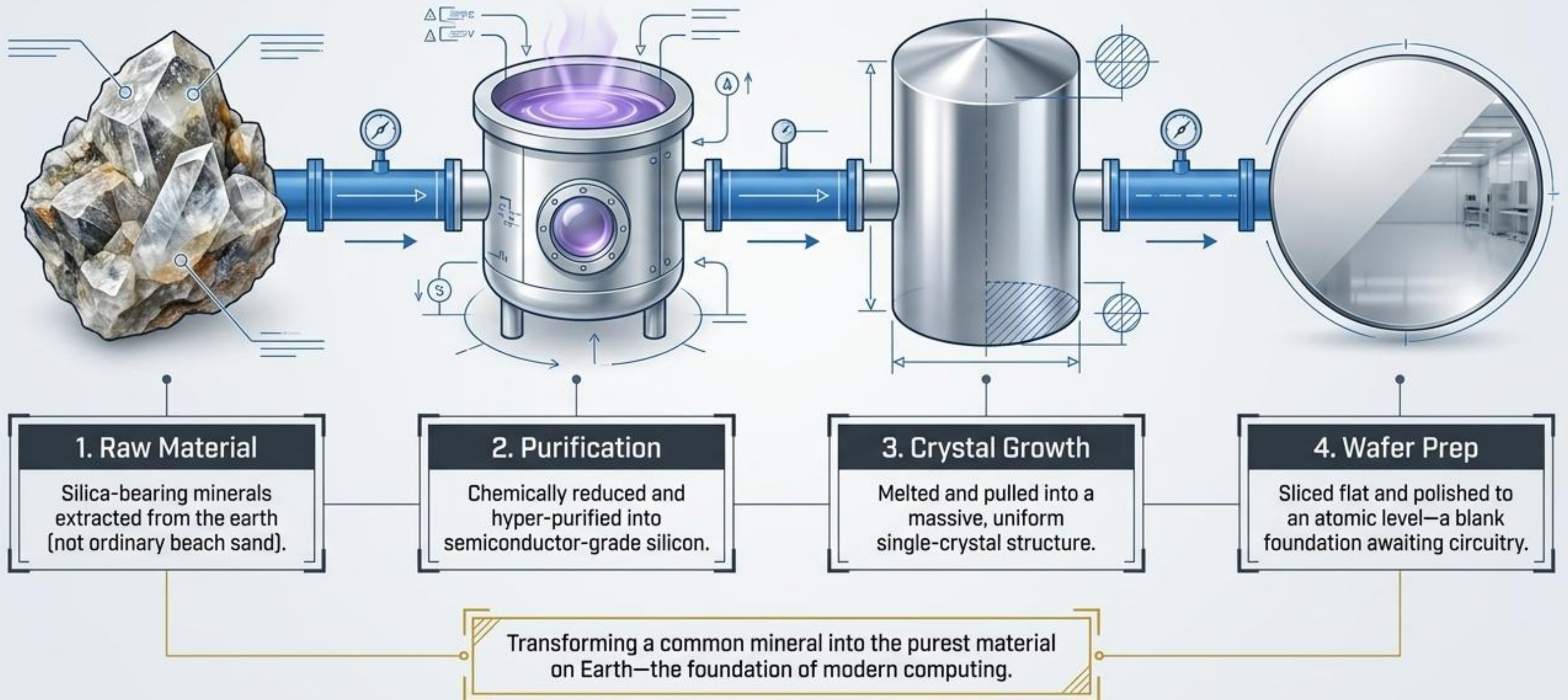


Semiconductor (Silicon)

Electrical behavior can be engineered to
create controllable microscopic switches.

Doping: Introducing carefully selected impurities allows engineers to precisely control the flow of electricity, turning silicon into a transistor.

From Raw Mineral to a Perfect Silicon Canvas

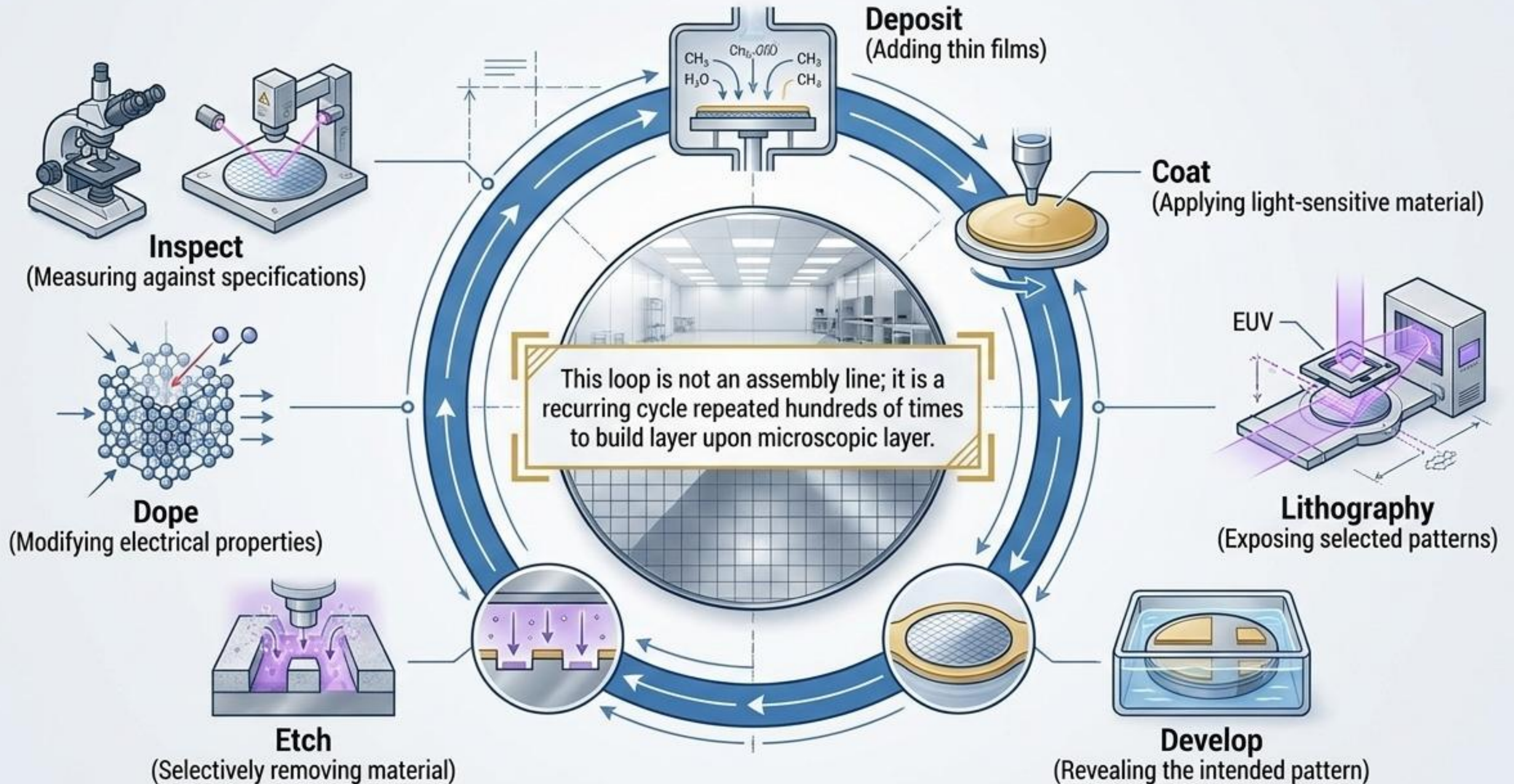




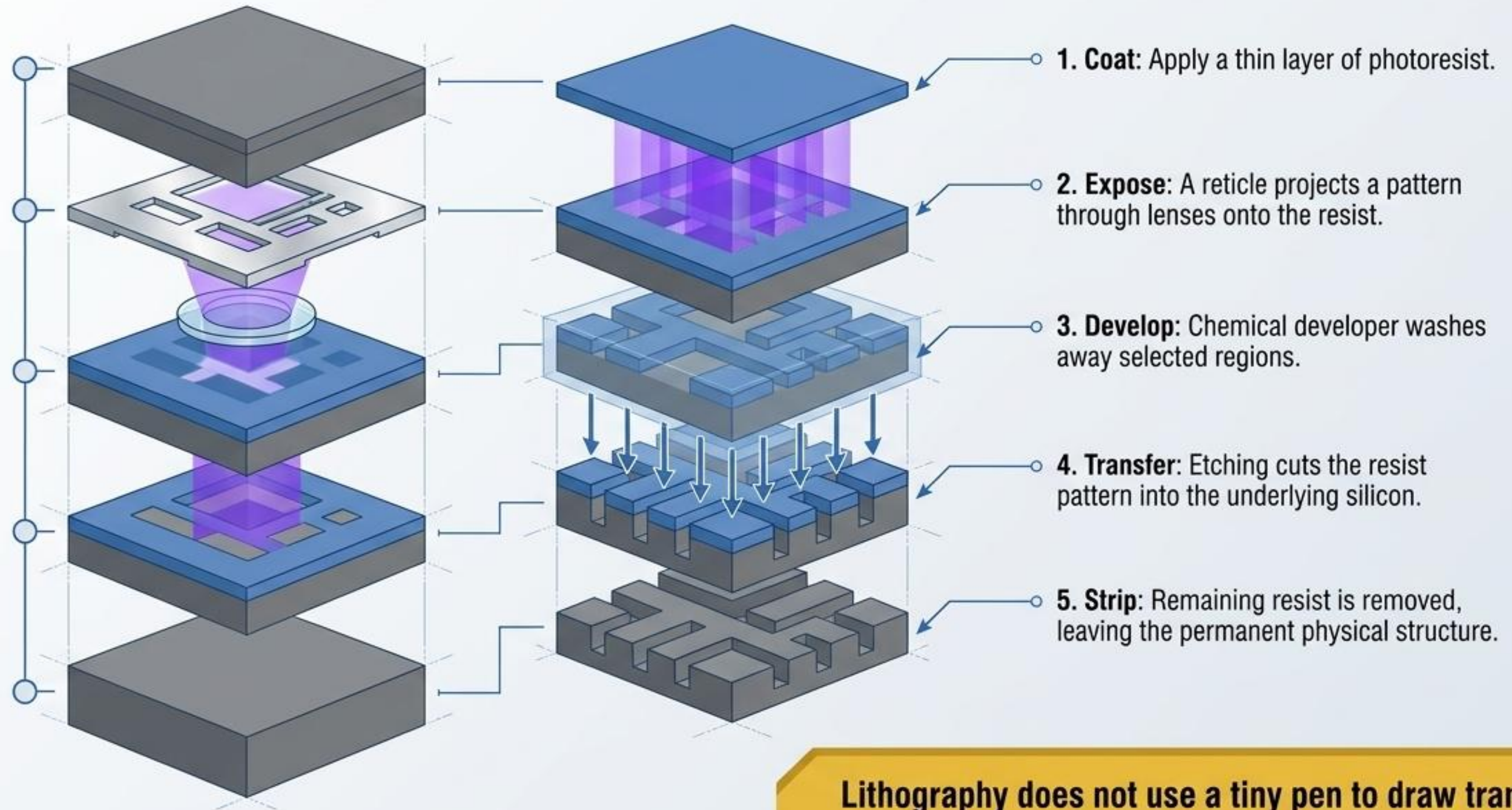
1. Purify → 2. Grow Crystal → 3. Slice & Polish

A hyper-pure foundation, not a finished chip.

Inside the Fab: The Microscopic Manufacturing Loop



Photolithography: Printing the Invisible



Lithography does not use a tiny pen to draw transistors; it uses light for optical pattern transfer.

Printing the Invisible



Coat



Expose



Develop

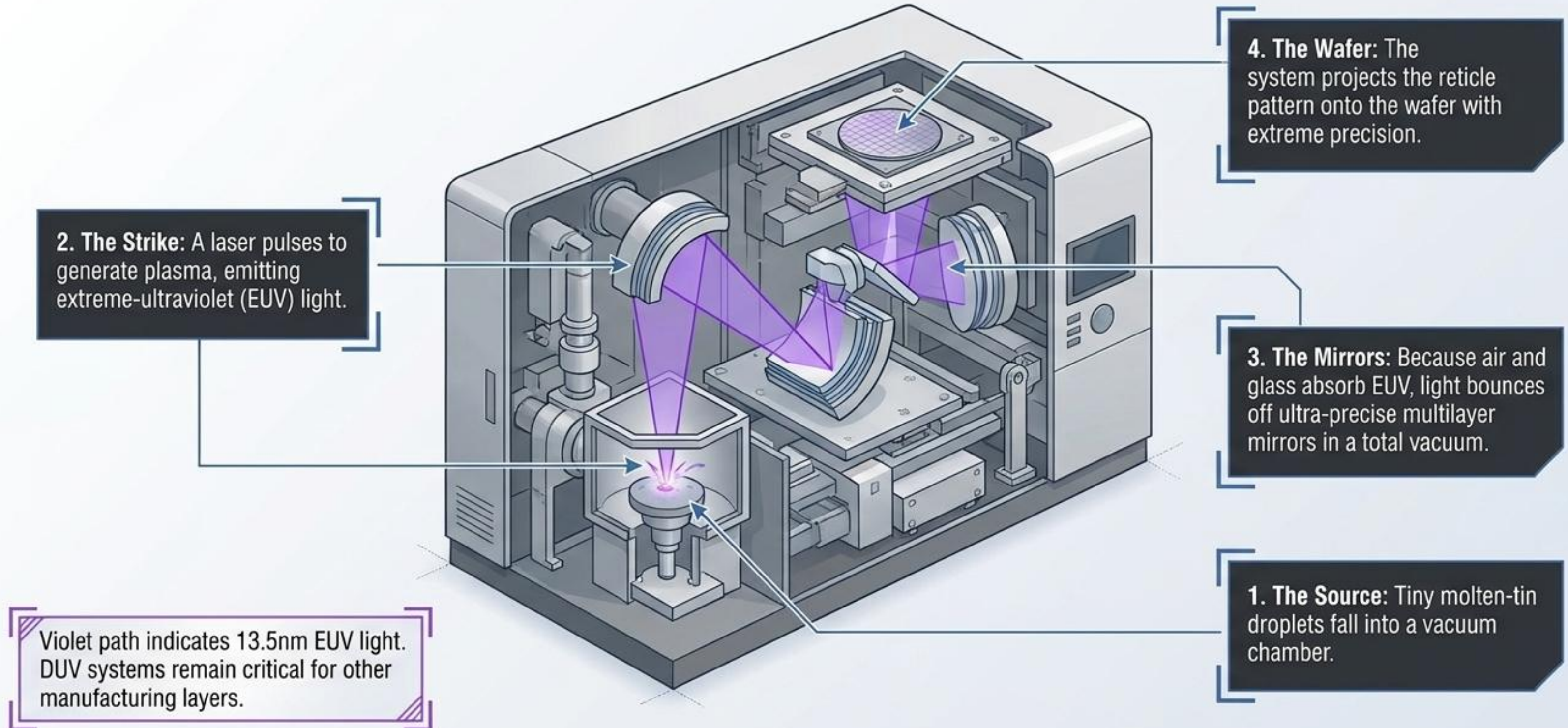


Etch



Strip

EUV Lithography: Engineering Light at 13.5 nm



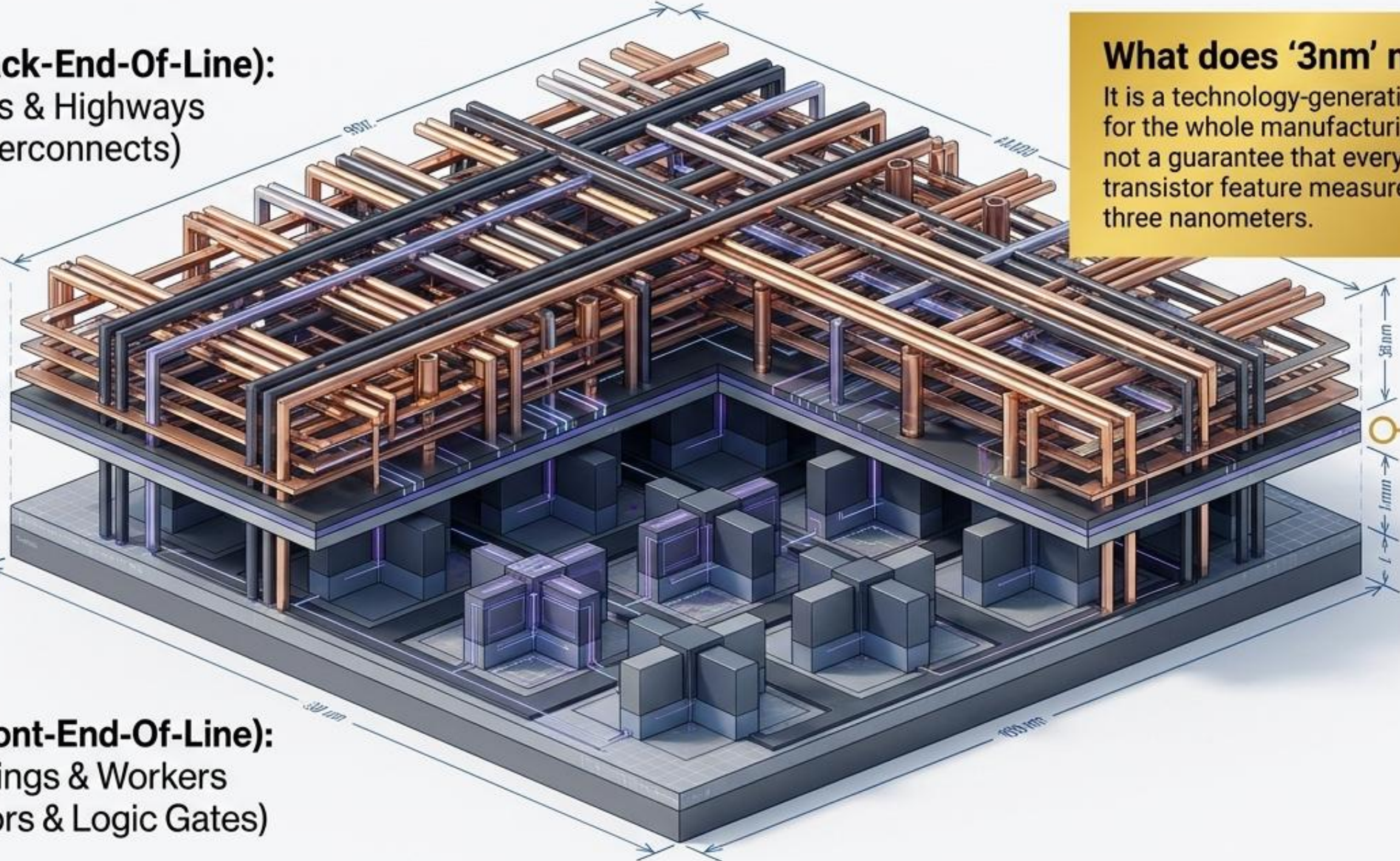
How Billions of Transistors Become a Processor

BEOL (Back-End-Of-Line):
The Roads & Highways
(Metal Interconnects)

What does '3nm' mean?

It is a technology-generation label for the whole manufacturing process, not a guarantee that every single transistor feature measures exactly three nanometers.

FEOL (Front-End-Of-Line):
The Buildings & Workers
(Transistors & Logic Gates)



The Hostile Nanoscale: Why Cleanrooms Exist



At the nanoscale, a single speck of dust acts like a meteor striking a city.

The Fab Environment

Airborne Particle Limits:

Filtered to levels millions of times cleaner than outside air.



Chemical & Vibration Control:

Strict regulation of humidity, temperature, and microscopic movement.



Automated Transport:

Robotic overhead tracks move wafers to prevent human contact.

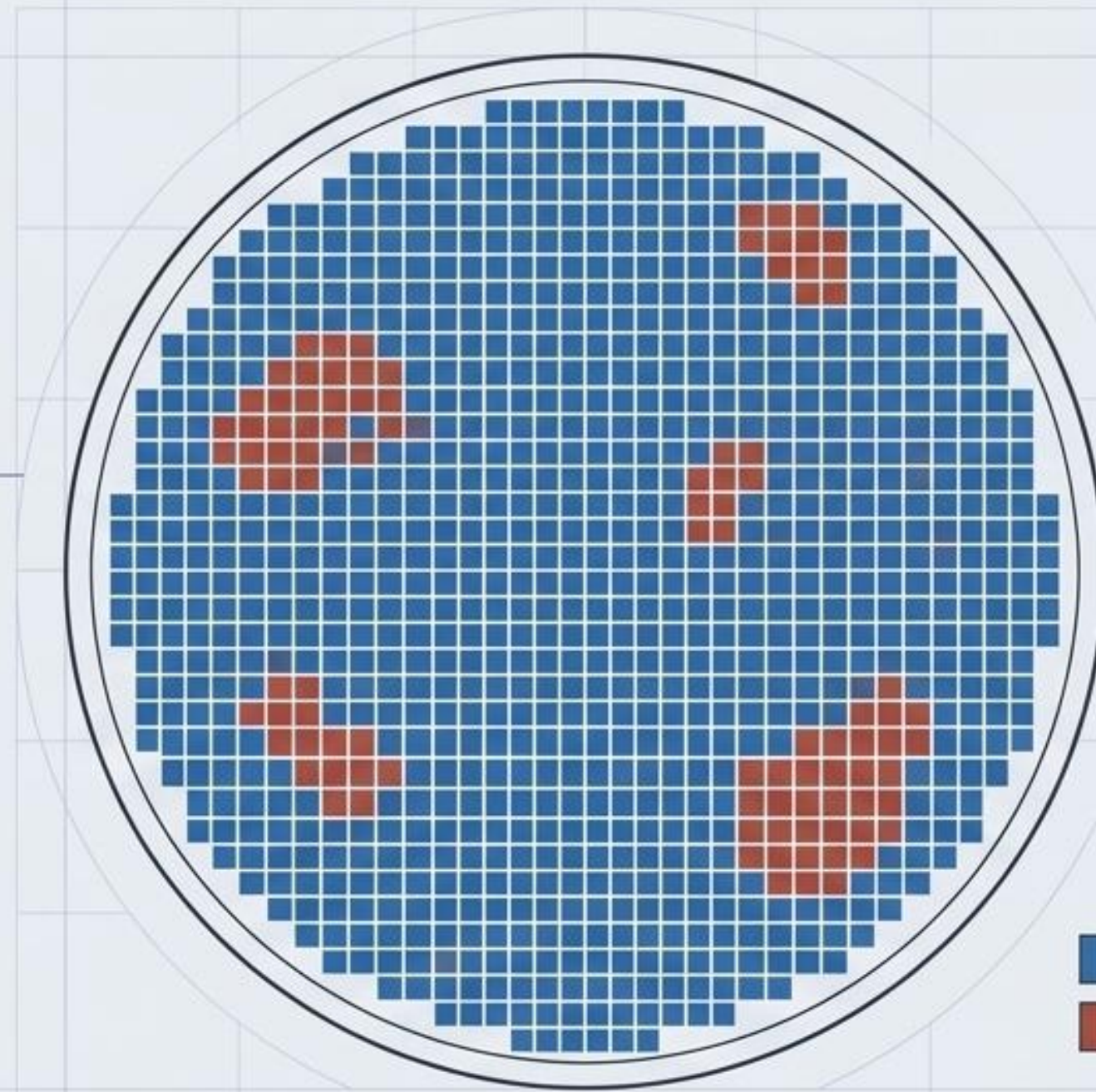


Human "bunny suits" are worn primarily to protect the microscopic chips from human contamination, not the other way around.

Yield: The Unforgiving Economics of Manufacturing

Yield:

The proportion of manufactured dies on a wafer that meet strict acceptance criteria and function perfectly.



Wafer Yield Map

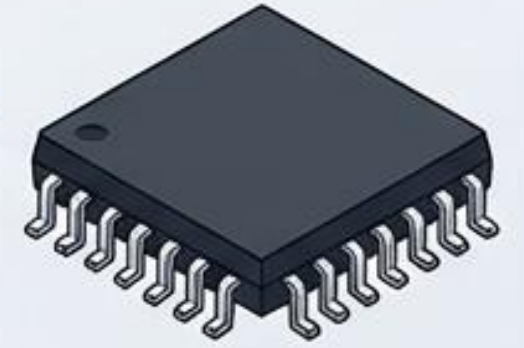
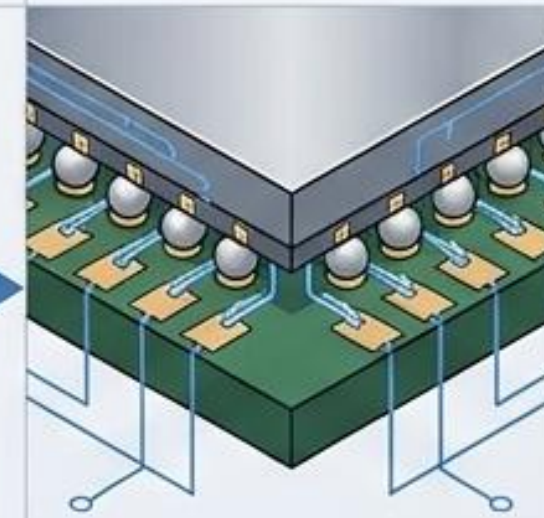
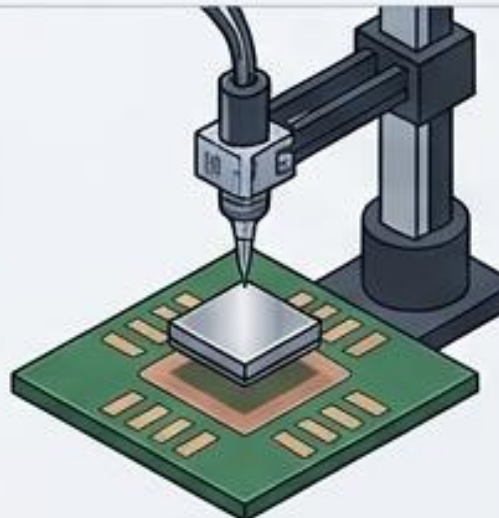
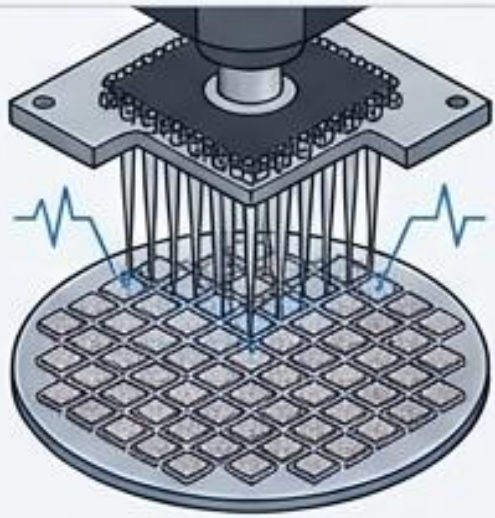
Blue: Functional Die
Red: Defective Die

The Economic Reality

The enormous cost of processing a wafer is locked in long before the factory knows exactly how many chips will survive testing.

A microscopic defect can render an entire die useless.

From Wafer to Finished Component



1. Wafer Probe:

Testing individual dies while on the wafer.

2. Dicing:

Precision saws separating the wafer into individual dies.

3. Die Attach:

Securing a single die to a package substrate.

4. Wiring/Bumps:

Creating microscopic electrical connections.

5. Encapsulation:

Sealed for thermal and physical protection.

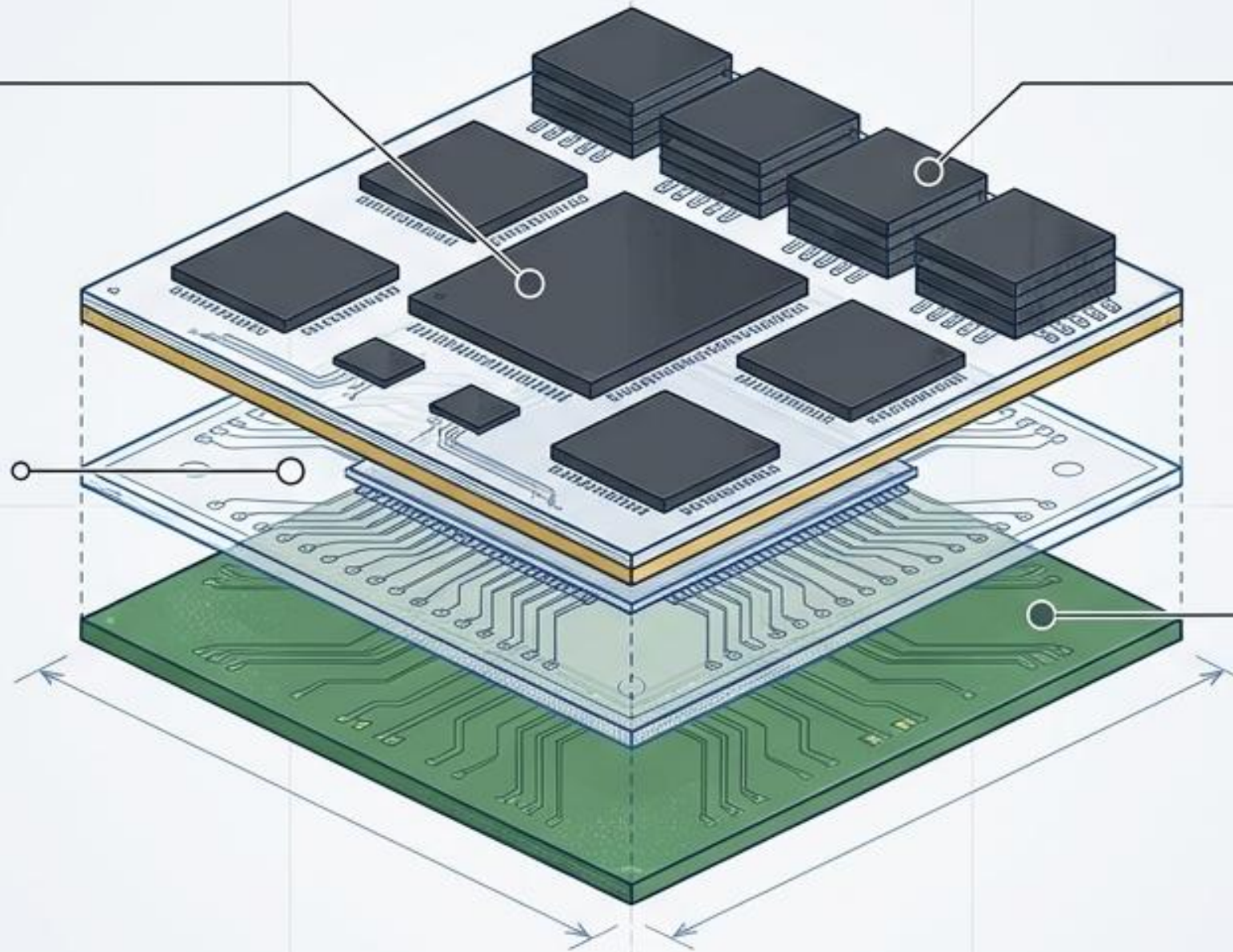
The Anatomy of a Modern AI Accelerator

Compute Dies (Logic): The processors performing calculations.

High-Bandwidth Memory (HBM): The tightly packed data pantry.

Silicon Interposer: The ultra-fast connection highway bridging memory and logic.

Substrate: The physical foundation that plugs into the server motherboard.



Takeaway: Advanced AI chips are highly integrated systems of multiple dies ('chiplets'), not just one monolithic piece of silicon.



1. Design
(Architecture)

2. Foundries
(Fabrication)

3. Equipment
(Machines & Chemicals)

4. OSAT
(Packaging & Test)

Who Actually Makes the Chips?

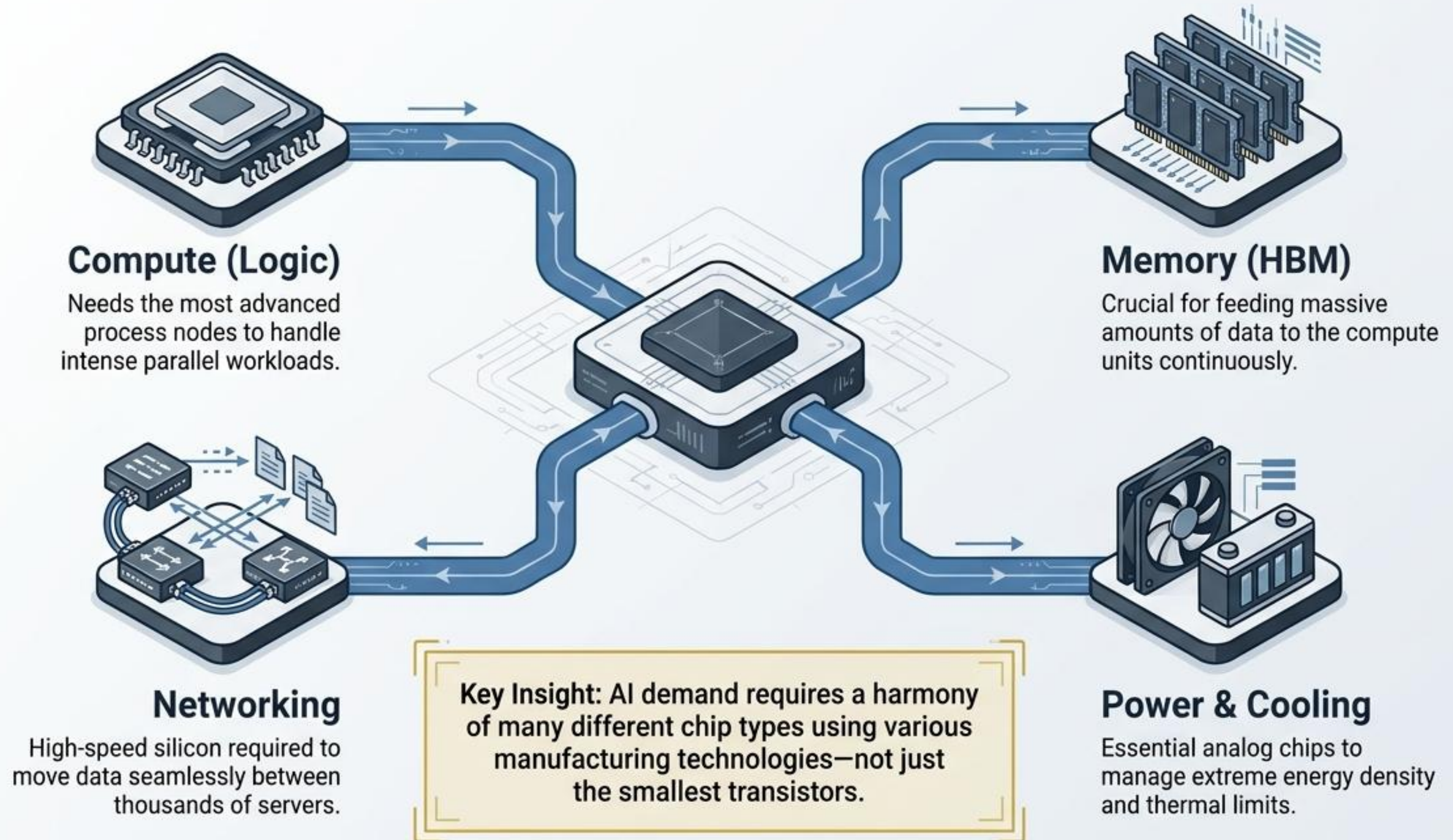
	Who Designs	Who Builds	Who Packages
Fabless Companies	✓ Designs architectures and circuits.		
Foundry		✓ Manufactures silicon dies for others.	
IDM (Integrated Device Manufacturer)	✓ Designs and fabricates in-house.	✓ Designs and fabricates in-house.	✓ Designs and fabricates in-house.
OSAT			✓ Assembly and testing of final components.

Equipment & Materials Providers

Enablers providing lithography tools, gases, and ultra-pure silicon to the entire ecosystem.

A disruption in any single node of this globally distributed supply chain delays the final product.

Why AI is Redefining Chip Demand



India's Semiconductor Ambitions



Data Overview (Source: Gov. Backgrounder, Aug 2026)

12 Approved
Projects

across **6** states

₹1.64 lakh crore

committed investments

Ecosystem Scope

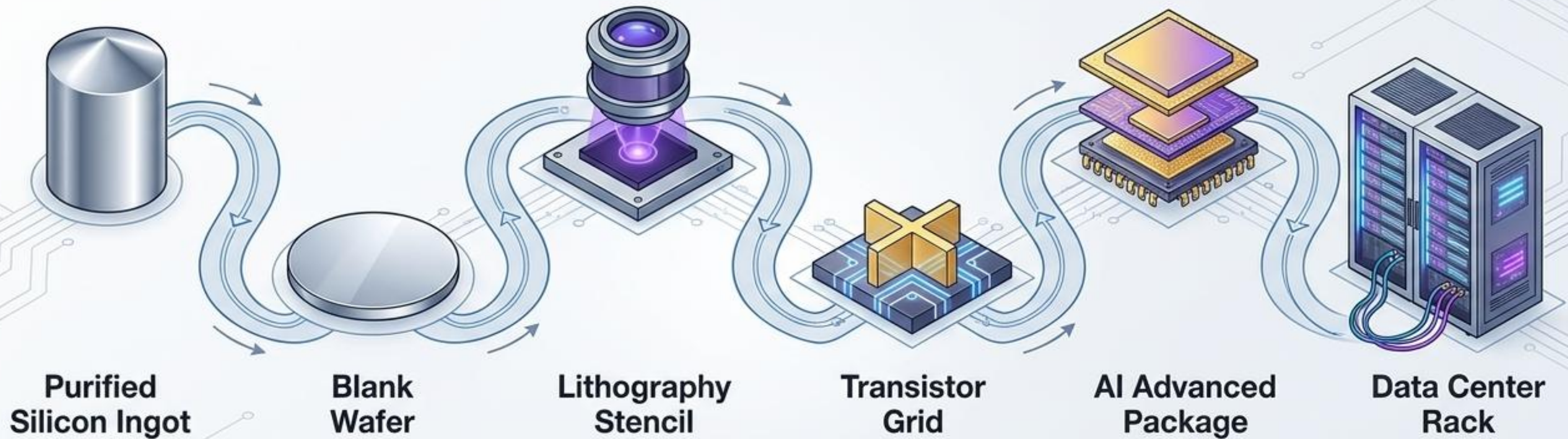
Projects span **silicon fabs**, compound semiconductors, display fabrication, and advanced packaging.



Crucial Context: Ecosystem vs. Production

Building a semiconductor ecosystem is a **multi-step journey** including **design, mature nodes, and packaging**. It is distinct from immediately producing leading-edge logic at global scale.

From Earth's Crust to the Future of Computing



A modern computer chip is the result of precision engineering across materials science, physics, and global collaboration. Its value comes not from the sand it started as, but the extraordinary process that turns it into the engine of the digital world.



1. Design
(Architecture)

2. Foundries
(Fabrication)

3. Equipment
(Machines & Chemicals)

4. OSAT
(Packaging & Test)



THANK YOU!

The future belongs to those who **build it continuously.**



START TODAY.
Take the first step.



STAY CONSISTENT.
Small steps, big results.



BUILD ONE PROJECT
at a time.



PRESENTED BY

DAKSH KHANDELWAL

FOUNDER, THINKHUB



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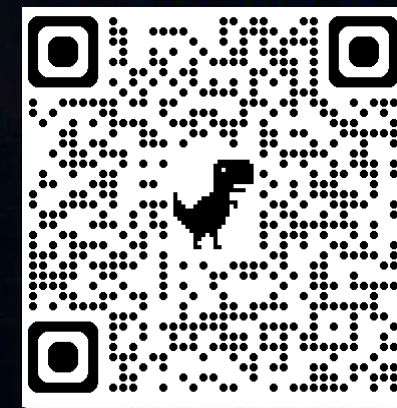


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KEEP LEARNING. KEEP BUILDING. KEEP GROWING.